HedgeProof

Protect city bus budgets with verifiable AI hedge research.

01 TEAM DETAILS

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Institution	JSS Science & Technology University Mysuru
Track & domain	Agentic AI · Quantitative finance / commodity hedging

02 Problem Statement

A city bus agency must budget for diesel before it knows the purchase price. Price spikes can put pressure on reserves and service budgets. Hedging is already a real transit-finance practice: MTA's October 2025 report describes hedging 50% of its next 12 months of projected diesel purchases [1].

Target user: the treasury analyst preparing a fuel-budget recommendation for the agency's finance committee. The hard question is whether a proposed hedge offsets the agency's actual price exposure, survives an honest backtest and stays within its approved limits.

A persuasive AI explanation is insufficient. A future-data leak, a mismatched futures contract or an oversized position can make an attractive backtest unsafe to act on.

03 Proposed Solution

HedgeProof turns an exposure forecast and a written policy into a bounded research task. Agents retrieve source evidence, propose a hedge, write and repair an estimator, challenge its assumptions and decide whether to revise or abandon it. Independent Python services calculate every result and control access to paper execution.

> **Core contribution: the Mandate-Bound Evidence Gate.**
> Every paper order must match a verified research record: the data, estimator, evaluation and policy versions. Missing evidence, altered artifacts or a breached rule block the order, with an explicit reason.

Research	Compare an adaptive hedge with unhedged and fixed 50% baselines. Investigate basis risk: the mismatch between local diesel and hedge prices.
Verify	Recompute budget-risk metrics; freeze one candidate before a single final holdout evaluation. A failed hedge remains a useful result.
Demonstrate	Show a traceable evidence packet, a policy veto and paper-order reconciliation in one dashboard.

> **Illustrative demo:** for 420,000 forecast gallons, a 50% hedge is five 42,000-gallon contracts [3]. A malicious suggestion to hedge 150% requests fifteen contracts; the independent gate rejects it against the demo's 75% cap. These are scenario inputs, not claimed results.

04 Technical Approach

One workflow, five bounded roles. A LangGraph state machine coordinates a shared LLM through typed tool calls [4]. It permits at most six candidate configurations and two code repairs per candidate; every attempt is logged.

Agent role	Autonomous action / tool
Research	Read cited sources and data profiles; propose an estimator and diagnostic plan.
Coder / Backtester	Write a restricted estimator function; run it in a sandbox; repair failed tests.
Critique	Request lag, subperiod and double-cost checks; revise, reject or stop research.
Risk	Interpret gate failures and propose smaller exposure; cannot change policy.
Execution	Submit eligible paper intents; query order status before retrying a timeout.

Quant engine: one agency, one price pair, one monthly cycle. Use EIA Gulf Coast diesel spot and NY Harbor futures [2]. Model one fuel purchase per month. On the first trading session, enter the second-nearby futures contract; exit the same delivery-month contract on the next month's first session, when it is front-month [2,3]. Features use only earlier available observations.

> **Estimator:** $h = \text{clip}(\text{Cov}(\Delta S, \Delta F) / \text{Var}(\Delta F), 0, 0.75)$
> **Position:** $n = \text{floor}(h \times Q / 42{,}000)$
> **Net fuel cost:** $Q \times S_{exit} - 42{,}000 \times n \times (F_{exit} - F_{entry}) + \text{costs}$
> Q: forecast gallons; S: spot price; F: matched futures contract. Δ denotes monthly price changes. Fit h on 24 or 36 completed months. Reject missing data or zero variance. Include stated fees and slippage; floor sizing preserves the cap.

Measure budget protection. Budget = Q × last available spot price, frozen before entry. Report the variance of cost minus budget, its empirical 95th percentile (Budget-at-Risk), and cash-stress exposure. Compare identical dates, volumes and costs across all baselines.

Layer	Technology and purpose
Agents & UI	Python, LangGraph, Pydantic, Claude API and Streamlit; typed state, bounded loops and a visible decision trace.
Quant & data	NumPy / pandas / SciPy / statsmodels; a small cash-flow engine, Parquet snapshots and SQLite trial records.
Trust boundary	Docker: no network, read-only past-data window, resource limits. Generated code returns estimates; trusted code computes metrics.
Gate & simulator	FastAPI service outside agent write access; policy checks, SHA-256 artifact hashes, transactional holdout lock and unique order IDs.

Integrity. Warm up and develop on 2014-17; validate in 2018-19. Purge boundary-crossing positions; reserve February 2020-March 2024 as the sealed holdout. Freeze one candidate and acceptance rules; evaluate once, with no retuning. Data cutoffs are enforced by the service, not agent judgment.

Accounting is separate. A stretch module reports one selected GASB-informed effectiveness diagnostic for the designated hedged portion [7]. It supports accountant review, not certification of accounting treatment, legal authority or future performance.

> **Hard gate:** fresh replay inputs; approved instrument; no net shorts; ≤75% of forecast gallons; one-month holding limit; cash-stress cap; matching evidence hashes; no duplicates. Unknown state halts. Paper release requires human approval.

05 Innovation

Our contribution links a city’s fuel exposure, executable policy and empirical evidence to each paper order. Multi-agent roles and code-calculated finance are established ideas.

Reference / baseline	HedgeProof’s specific distinction
TradingAgents [5]	Trading-firm agent roles inform decisions; our objective is fuel-budget protection with an independent policy gate.
FinRobot [6]	Already separates deterministic finance from LLM narration. We apply that principle to exposure-bound hedge research and paper-order eligibility.
Fixed hedge script	Tests whether adaptive investigation catches more flaws while accepting sound cases.

06 Feasibility & Impact

Hours	Build milestone and ownership
0-2	All: verify data / contract mapping; freeze exposure, policy, costs, splits and acceptance thresholds.
2-8	Saicharan: graph / ledger. Nikshep: quant engine. Kruthik: agents / sandbox. Lohith: gate / simulator / UI. First end-to-end slice by hour 8.
8-18	Integrate critique, sealed evaluation and timeout recovery; run baselines and fault cases.
18-24	Fix, reproduce and document; rehearse and freeze the demo.

Evaluation targets, not results: 16 predefined cases, eight valid and eight faulty: future-data use, sign error, stale data, holdout reuse, oversizing, short exposure, duplicate intent and injected instructions. Targets: zero policy-violating or duplicate orders; ≥7/8 faults caught; ≥7/8 valid cases accepted.

Compare a single agent with equal tools / budget and a fixed script; ablate the gate. Report risk with sample counts and block-bootstrap intervals, false rejections, reproducibility, calls and runtime. Freeze the pass rule: lower budget-error variance and no worse Budget-at-Risk than unhedged, with no rule breach; otherwise abstain.

Three-minute demo: research → estimator run → future-data attempt blocked → 150% order vetoed → sealed evaluation → approved paper order reconciled after a timeout. A failed hedge ends with an evidenced refusal.

MVP limits. Historical paper research; declared gallon forecast; spot-price proxy for invoices. EIA futures history ends 5 April 2024 [2]. Fills, release lags and cash stress are assumptions; revisions remain a limitation. No live broker or paid feed is required.

Impact. Reproducible evidence for transit-budget decisions. Next: an invoice-based shadow pilot with licensed contract data and treasury review. Four members build the prototype during the hacking period. No performance or savings are claimed.

Sources & acknowledgments

[1] MTA, Fuel Hedge Program Report, 27 Oct 2025.
[2] EIA: spot prices; NYMEX futures; contract definitions.
[3] CME Group, NY Harbor ULSD futures, Rulebook chapter 150.
[4] LangChain, LangGraph overview.
[5] TauricResearch, TradingAgents repository.
[6] AI4Finance Foundation, FinRobot repository.
[7] Georgia SAO, Derivative Instruments, revised 11 Apr 2012.
Checked 19 Oct 2026. Planned public data, libraries and LLM API use will be attributed under applicable terms. Proposal drafted with AI assistance.